

| Edge detection via Hough Transformations

1) Remove noise in image if any

→ It's a voting technique used to detect imperfect instances of objects within a certain class of shapes

lines: from cartesian space $(x, y) \rightarrow (p, \theta)$

why not $y = mx + c$?

$$p = x \cos \theta + y \sin \theta$$



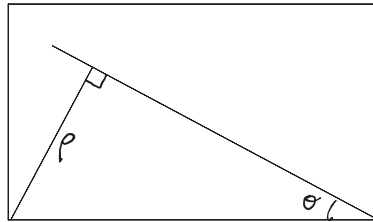
$\perp$ distance from origin to the line

θ angle formed by $\perp$ line & horizontal axis

Hough transform

→ Uses a voting procedure → param space

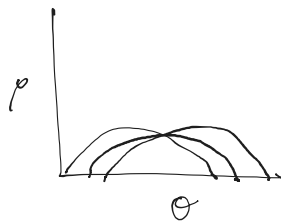
→ Possible lines are found by identifying local maxima in this accumulator space



- A point (x_i, y_i) can have infinite no. of lines passing through it.

in parametric space $(p, \theta) = 1$ sinusoidal curve

- collinear lines will intersect at a common point



Pseudocode:

- create a voting tracker = []
- Parameter space : $\theta \rightarrow 0, 180$
 $p \rightarrow 0, \text{diagonal}$

→ diagonal len = $\sqrt{w^2 + h^2}$

Initialize store = []

→ calc → $\text{math.cos}()$

$$p = x \cdot \cos(\theta) + y \sin(\theta)$$

→ rho-index = $\text{int}(\text{rho}) + \text{diag len} - \text{if -ve}$

hough_peaks ()

If votes > threshold $\rightarrow$ edge

real_rho = r_idx - diag len

real_theta = t_idx

$\hookrightarrow$ out [rho, theta]

math.ceil $\rightarrow$ smallest integer $\geq x$

Affine Transformations

Rotation and Skew

Rotation

$p_{(i,j)}$ i - Row no.
 j - col no.

$\rightarrow$ We are moving the pix to a new address (i', j')

$$\begin{bmatrix} i' \\ j' \end{bmatrix} = \begin{matrix} \text{Transformation} \\ \text{matrix} \end{matrix} \times \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} \cos(\phi) & \sin(\phi) & 0 \\ -\sin(\phi) & \cos(\phi) & 0 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

new row, $i' = i \cdot \cos(\phi) + j \cdot \sin(\phi)$

new col, $j' = -i \cdot \sin(\phi) + j \cdot \cos(\phi)$

The origin is at (0,0) - Left-top corner.

$\downarrow$
rotate
 $\downarrow$
shift back to screen

Skew transformation

$\rightarrow$ Column j stays same, while row i shifts up or down depending on how far right you are

$$\begin{bmatrix} i' \\ j' \end{bmatrix} = \begin{bmatrix} 1 & \tan(\phi) & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

new row: $i' = i + \tan(\phi) \cdot j$

new col: $j' = j$